

REPORT

Title

Implementation of the Travelling Salesman Problem Using Python

Aim

To implement the Travelling Salesman Problem in Python and determine the shortest route that visits every city exactly once and returns to the starting city.

Problem Statement

The Travelling Salesman Problem (TSP) is an optimization problem in which a salesman must travel through a set of cities, visiting each city exactly once before returning to the starting city. The objective is to identify the route with the minimum total travel distance or cost. The program should compute the optimal path for any given distance matrix and display the shortest route along with its minimum cost.

Requirements

- Python 3.x

Procedure

1. Represent the distances between cities using a distance matrix.
2. Select the starting city.
3. Generate all possible routes that visit each city exactly once.

4. Compute the total distance for every possible route.
5. Include the distance required to return to the starting city.
6. Compare all route costs.
7. Identify the route with the minimum total distance.
8. Display the shortest route and its minimum travel cost.

Program

```
from itertools import permutations

def travelling_salesman(dist, start):
    cities = list(range(len(dist)))
    cities.remove(start)
    min_cost = float('inf')
    best_path = []
    for route in permutations(cities):
        cost = 0
        current = start

        for city in route:
            cost += dist[current][city]
            current = city

        cost += dist[current][start]

        if cost < min_cost:
            min_cost = cost
            best_path = [start] + list(route) + [start]

    return best_path, min_cost
```

```
distance = [  
    [0, 10, 15, 20],  
    [10, 0, 35, 25],  
    [15, 35, 0, 30],  
    [20, 25, 30, 0]  
]  
  
path, cost = travelling_salesman(distance, 0)  
print("Shortest Path:", path)  
print("Minimum Cost:", cost)
```

Output

```
Shortest Path: [0, 1, 3, 2, 0]  
Minimum Cost: 80
```

Result

The Travelling Salesman Problem was successfully implemented in Python. The program evaluated all possible routes and identified the route with the minimum travel cost.

Conclusion

The Travelling Salesman Problem demonstrates the application of optimization and search techniques in Artificial Intelligence. By evaluating possible routes and selecting the one with the minimum total cost, the algorithm provides an optimal solution for small-sized problems. TSP has practical applications in logistics, transportation, delivery systems, manufacturing, and network routing.

